LAB 9: Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES:

1. To understand the working mechanism of a Wireless Local Area Network (WLAN) and the configuration of a Wireless Router.
2. To configure Service Set Identifier (SSID) and WPA2 security protocols to secure the wireless network.
3. To establish connectivity between client laptops and the Wireless Router.
4. To verify wireless communication using packet analysis tools (Ping).

THEORY:

1. WLAN (Wireless Local Area Network)

A WLAN is a wireless computer network that links devices using radio communication within a limited area such as a home, school, or office.

Operation:
WLAN uses high-frequency radio waves (2.4 GHz or 5 GHz) to transmit data between client devices and a wireless router or access point. It eliminates the need for physical Ethernet cables.

Advantages:

- Mobility and flexibility
- Easy installation
- Reduced cabling cost

2. SSID (Service Set Identifier)

SSID is the unique name assigned to a wireless network.

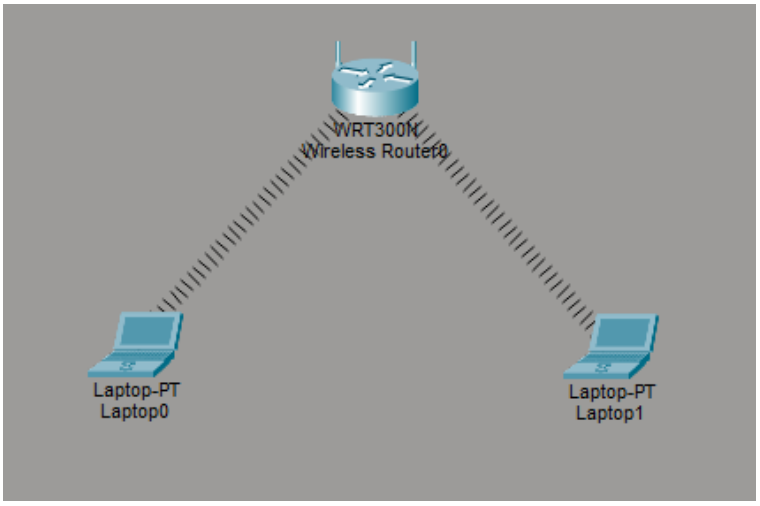
Operation:
Client devices scan available wireless networks and identify them using their SSID. Users select the correct SSID to connect to the intended network.

3. WPA2 (Wi-Fi Protected Access 2)

WPA2 is a wireless security protocol used to protect data transmission.

Operation:
WPA2-Personal uses AES encryption and requires a Pre-Shared Key (PSK) for authentication. Only users with the correct passphrase can connect to the network.

NETWORK TOPOLOGY:



- One **Wireless Router (WRT300N named Wireless Router0)**
- Two client devices (**Laptop0 and Laptop1**)

The wireless router acts as the central access point, broadcasting the wireless signal to both laptops.

IP CONFIGURATION TABLE:

Device	Interface	IP Address	Subnet Mask	Default Gateway	Allocation Method
Wireless Router0	LAN/Wireless	192.168.0.1	255.255.255.0	-	Static
Laptop0	Wireless0	192.168.0.102	255.255.255.0	192.168.0.1	DHCP
Laptop1	Wireless0	192.168.0.103	255.255.255.0	192.168.0.1	DHCP

The Wireless Router was configured to act as a DHCP server, automatically assigning IP addresses to connected clients.

CONFIGURATION PROCEDURE:

1. Router Setup

- Accessed the GUI tab of the Wireless Router.
- Under **Wireless Settings**, the SSID was set to:

```
HCOE
```

- Under **Wireless Security**, the following settings were configured:
    - Security Mode: WPA2-Personal
    - Encryption: AES
    - Pre-Shared Key:

```
hcoe@123
```

2. Client Connection

On Laptop0:

- Opened the **PC Wireless** application.
- Navigated to the **Connect** tab.
- The system scanned for available networks.
- The SSID **HCOE** was detected with strong signal strength.
- Selected the network and clicked **Connect**.

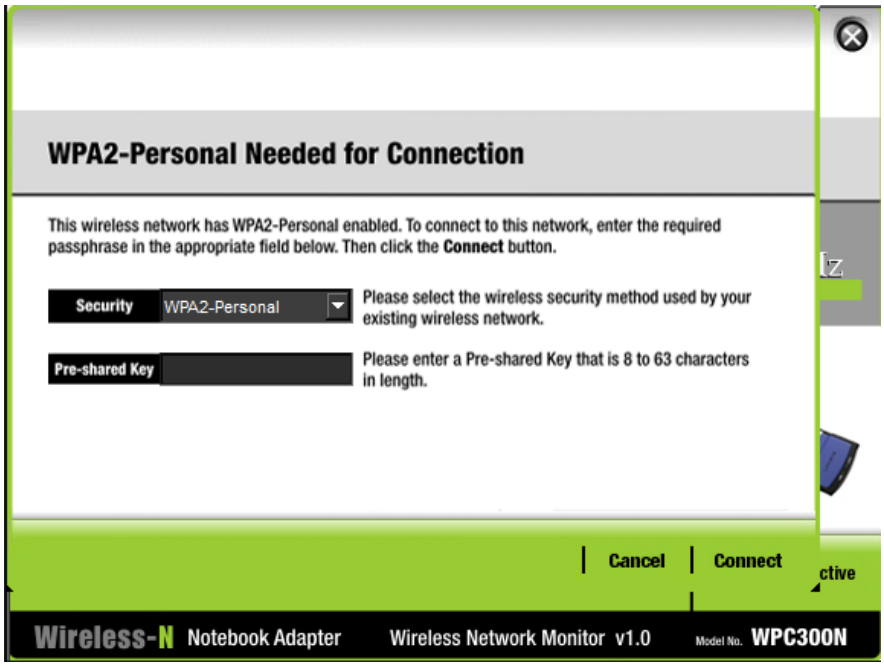
3. Authentication

A security prompt appeared requesting the Pre-Shared Key.

- Entered:

```
hcoe@123
```

- Authentication was successful.



4. IP Assignment

Once connected:

- Laptops automatically requested IP addresses from the router via DHCP.
- IP addresses were successfully assigned (192.168.0.102 and 192.168.0.103).

TESTING:

Connectivity Verification

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:

Reply from 192.168.0.103: bytes=32 time=46ms TTL=128
Reply from 192.168.0.103: bytes=32 time=20ms TTL=128
Reply from 192.168.0.103: bytes=32 time=63ms TTL=128
Reply from 192.168.0.103: bytes=32 time=53ms TTL=128

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 63ms, Average = 45ms
```

This confirms proper wireless communication between the devices.

RESULT:

The wireless network was successfully configured.

- SSID was broadcast correctly.
- WPA2 security prevented unauthorized access.
- Laptops successfully authenticated using the Pre-Shared Key.
- DHCP automatically assigned IP addresses.
- Ping test verified successful wireless communication.

DISCUSSION & CONCLUSION:

During this lab session, a wireless network was successfully established using a WRT300N wireless router. The SSID "HCOE" was configured, and the network was secured using WPA2-Personal encryption with a pre-shared key.

We observed that the client laptops were able to detect the broadcasted SSID and authenticate using the correct credentials. After authentication, the devices obtained dynamic IP addresses from the router’s DHCP service.

Connectivity was verified using the ping command, which showed 0% packet loss. This confirms that the wireless configuration, security settings, and DHCP functionality were properly implemented.

Hence, the lab was completed successfully with proper knowledge and implementation of Wireless Network Setup and Packet Analysis using Cisco Packet Tracer.